$n_1 = n_2 = 5$ $T_1 = 2.5 + 5 + 7.5 + 9 + 10 = 34$.

$T_1^* = n_1(n_1+n_2+1) - T_1 = 5 \times 11 - 34 = 21$ $T = T^* = 21$.

DATA SET DS1502

13. Eye Movement In an investigation of the visual scanning behavior of deaf children, measurements of eye movement were taken on nine deaf and nine hearing children. The table gives the eye-movement rates and their ranks (in parentheses). Does it appear that the distributions of eye-movement rates for deaf children and hearing children differ?

	Deaf Children	Hearing Children
	2.75 (15)	.89 (1)
	2.14 (11)	1.43 (7)
	3.23 (18)	1.06 (4)
	2.07 (10)	1.01 (3)
	2.49 (14)	.94 (2)
	2.18 (12)	1.79 (8)
	3.16 (17)	1.12 (5.5)
	2.93 (16)	2.01 (9)
	2.20 (13)	1.12 (5.5)
Rank Sum	126	45

H_0: the distribution of eye-movement rates for deaf children and hearing children are same

H_A: The distribution of eye-movement rates for deaf children and hearing children are different.

$n_1 = 9,\ n_2 = 9,\ T_1 = 45$ $T^* = n_1(n_1+n_2+1) - T_1 = 9 \times 19 - 45 = 126$

$T = \min(T_1, T^*) = 45$

$T_{0.025} = 62$.

$RR\ \{T \mid T \le 62\}$

$45 < 62 \Rightarrow$ Reject H_0

There's sufficient evidence to say that the distributions of eye-movement rates for hearing children is to the left of the deaf children.

smaller s^2 0.013

$\Rightarrow$ assume $\sigma_1^2 = \sigma_2^2$

$$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2} = \frac{4 \times 0.025 + 4 \times 0.013}{8} = 0.019$$

$H_0: \mu_1 - \mu_2 = 0$; $H_a: \mu_1 - \mu_2 > 0$

$\alpha = 0.05$

$$t_{STAT} = \frac{(\bar{X}_1 - \bar{X}_2) - 0}{\sqrt{s^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim t_{n_1+n_2-2}$$

$$t^* = \frac{5 - 4.86}{\sqrt{0.019\left(\frac{1}{5} + \frac{1}{5}\right)}} = 1.606$$

RR: $t_{8,0.05} = 1.86$

$\{t \mid t > 1.86\}$

$1.606 < 1.86 \Rightarrow$ non-reject H_0

There's no sufficient evidence to say that population 1 distribution is to the right of population 2 distribution.

result same as part a.